

Computer Science & Information Technology

Computer Networks

DPP: 1

Error Control

- Q1** Consider ASCII character 'B' (ASCII value = 66) is transmitted by transmitter, but ASCII character 'A' (ASCII value = 65) is received by receiver. Identify the type of error ?
 (A) No any error
 (B) Single bit error
 (C) Burst Error
 (D) Data insufficient
- Q2** Consider ASCII character 'A' (ASCII value = 65) is transmitted by transmitter, but ASCII character 'T' (ASCII value = 84) is received by receiver. Count the number of corrupted bits?
- Q3** Let suppose, even parity is used in single-bit parity error detection technique. If transmitter finds total 91 one's in the data (excluding parity) then what should be parity bit value set by the transmitter ?
 (A) 0
 (B) 1
 (C) Can be any 0 or 1
 (D) Data insufficient
- Q4** Let suppose, even parity is used in single-bit parity error detection technique. If receiver find total 93 one's in the received block (including parity) then what receiver concluded?
 (A) No any error detected
 (B) Error detected
 (C) Unable to detect error
 (D) Data insufficient
- Q5** Consider single-bit parity error detection technique, the number of data bits are 6 (excluding parity).
 Count the number of valid code words?
- Q6** Consider the degree of generator polynomial function is n , what should be length of divisor (in bits)?
 (A) n (B) $n + 1$
 (C) $n - 1$ (D) $2n$
- Q7** Consider generator polynomial function is $x^3 + 1$, if data is 1011010110 then calculate CRC ?
 (A) 001 (B) 011
 (C) 101 (D) 110
- Q8** Consider generator polynomial function is $x^3 + x + 1$, if received codeword by receiver is 1101011010100 then what receiver concluded ?
 (A) No any error detected
 (B) Error detected
 (C) Unable to detect error
 (D) Data insufficient
- Q9** Identify correct statement(s) regarding CRC error detection technique.
 (A) CRC can detect any length burst error up-to the degree of generator polynomial function $G(X)$
 (B) If $(1 + X)$ is a factor of generator polynomial function $G(X)$ then CRC can detect all odd number bits error.
 (C) If generator polynomial function $G(X)$ does not divide $1 + X^k$, for any k upto frame length



then CRC can detect any two bit error.

(D) To ensure correct operation of CRC the generator polynomial function $G(X)$ should not be completely divisible by X .

- Q10** Consider 2D parity scheme for error detection, let suppose size of data block matrix (including parity bits) is 5×6 . Count total number of data bits in the data block (excluding parity bits) ?
- Q11** Consider 2D parity scheme for error detection with even parity is used and size of matrix is 4×5 . Let suppose 20 bits block (contains 12 data bits) is $0 \times A36EF$ (represented in hexadecimal) received by receiver. The received block contains n corrupted bits. What should be minimum possible value of n ?
- Q12** In hamming code, how many minimum number of parity bits required for 12 data bits?

Q13 Consider even parity in hamming code and first parity is placed at least significant bit position, the data bits 1101 is encoded as:

- (A) 1101110 (B) 1100100
(C) 1100110 (D) 1101101

Q14 Consider error control method has code words with 'd' hamming distance, receiver can detect upto _____bits errors.

- (A) d (B) $d - 1$
(C) $d + 1$ (D) $d/2$

Q15 To correct upto 8 bits error minimum hamming distance should be _____.

Q16 Consider an error control method has the following code words :

00000000, 00001111, 11110000, 11111111

What is maximum number of bit errors that can be corrected ?



Answer Key

Q1 (C)
Q2 3~3
Q3 (B)
Q4 (B)
Q5 64~64
Q6 (B)
Q7 (D)
Q8 (B)

Q9 (A, B, C, D)
Q10 20~20
Q11 1~1
Q12 5~5
Q13 (C)
Q14 (B)
Q15 17~17
Q16 1~1



Hints & Solutions

Q1 Text Solution:

Transmitter Data = B = 66

Received Data = a = 65

66 = 01000010

65 = 01000001

Q2 Text Solution:

Transmitter Data = A = 65 = 01000001

Received Data = T = 84 = 01010100

Number of corrupted bits = 3

Q3 Text Solution:

Single-bit parity, [Even parity]

Hence option B is correct.

Q4 Text Solution:

Single bit parity, even parity

Q5 Text Solution:

No. of data bits = 6

single-bit parity

No. of bits in the codeword = $(6 + 1) = 7$

No. of valid codewords = $2^6 = 64$

Q6 Text Solution:

degree $[G(x)] = n$

$G(x) = x^n + \dots + 1$

Q7 Text Solution:

```

1001 | 1011010110000
      1001
      ---
        1001
        1001
        ---
          1100
          1001
          ---
            1010
            1001
            ---
              110
              CRC
  
```

$G(x) = X^3 + 1$

Divisor = 1001

Q8 Text Solution:

```

1011 | 1101011010100
      1011
      ---
        1100
        1011
        ---
          1111
          1011
          ---
            1001
            1011
            ---
              1001
              1011
              ---
                1001
                1011
                ---
                  1010
                  1001
                  ---
                    11
  
```

$G(x) = x^3 + x + 1$

Divisor = 1011

if remainder at received is non-zero
then error detect

Q9 Text Solution:

All option are True

Q10 Text Solution:

2D parity

Block size = 5×6

No. of data bits = $(5 - 1) \times (6 - 1)$
= $4 \times 5 = 20$

Q11 Text Solution:

Received data = 0xA36EF (Even parity)

10100011011011101111

1 0 1 0 0

0 1 1 0 1

1 0 1 1 1

0 1 1 1 1

Min^m one bit Error

Ans = 1



Q12 Text Solution:

No. of data bits(m) = 12

No. of parity bits(r) = ?

$$2^r > (m + r)$$

Min^m value for $r = 5$

Q13 Text Solution:

Data = 1101, Even parity

$$\frac{1}{7} \quad \frac{1}{5} \quad \frac{0}{5} \quad \frac{0}{4} \quad \frac{1}{3} \quad \frac{1}{2} \quad \frac{0}{1}$$

Q14 Text Solution:

$$\text{Min}^m H \cdot D = d$$

It can detect upto $(d - 1)$ bits errors

It can correct upto $\left\lfloor \frac{d-1}{2} \right\rfloor$ bits error

Q15 Text Solution:

To correct 8 bits error min^m

$$H \cdot D = (2 \times 8 + 1) = 17$$

Q16 Text Solution:

$$\text{Min}^m H \cdot D (d) = \text{Min} [d(c_1, c_2), d(c_1, c_3), d(c_1, c_4),$$

$$d(c_2, c_3), d(c_2, c_4), d(c_3, c_4)]$$

$$= \min[4, 4, 8, 8, 4, 4]$$

$$d = 4$$

max^m no. of bits that can be corrected =

$$\left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{4-1}{2} \right\rfloor = 1$$



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